

● RAG Learning

Topic 6: Plant Structures and Their Functions

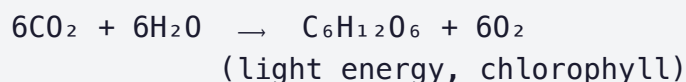
Edexcel GCSE (9–1) Biology (1BIO) — Paper 2

Plants are the foundation of almost every food chain on Earth. They capture light energy and use it to build glucose from the simple molecules carbon dioxide and water — a process called **photosynthesis**. From that glucose, plants make every other biological molecule they need: starch for storage, cellulose for cell walls, lipids for seed stores, and, with nitrogen from the soil, amino acids and proteins. In this topic you will learn how the photosynthesis reaction works, how environmental factors limit its rate, how leaves are perfectly adapted as photosynthetic organs, how water and food are transported through xylem and phloem, and how plant hormones such as auxins, gibberellins and ethene control growth and are used in agriculture.

6.1 Photosynthesis

Photosynthesis is the process by which green plants (and algae) use light energy to convert carbon dioxide and water into glucose and oxygen. It is an **endothermic** reaction — energy is taken in from the surroundings (in the form of light).

The overall balanced equation for photosynthesis is:

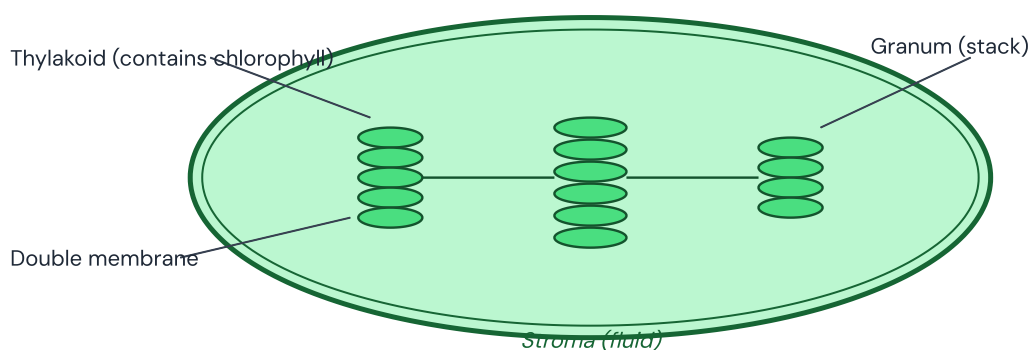


In words: six molecules of carbon dioxide react with six molecules of water to produce one molecule of glucose and six molecules of oxygen. Light energy is absorbed by the green pigment **chlorophyll**, which is found in **chloroplasts** — disc-shaped organelles in the cells of leaves (mainly the palisade mesophyll).

Feature	Detail
Reactants	Carbon dioxide (from air, through stomata) and water (from soil, via roots and xylem)
Products	Glucose ($C_6H_{12}O_6$) and oxygen (O_2)
Energy source	Light (usually sunlight)
Catalyst / absorber	Chlorophyll inside chloroplasts
Type of reaction	Endothermic — takes in energy from the surroundings
Site in plant	Mostly the palisade mesophyll cells of leaves

Exam Tip: You must be able to recall the word and symbol equations, state that photosynthesis is **endothermic**, and identify chlorophyll/chloroplasts as the site of light absorption. "Endothermic" means energy (light) is absorbed — the reverse (respiration) is exothermic.

Chloroplast — site of photosynthesis



6.2 Stages of Photosynthesis

Photosynthesis takes place in **two stages**. You only need to know them at the level below — Edexcel does *not* require the full Calvin cycle or detailed biochemistry.

Stage 1 — The Light-Dependent Stage

- Takes place in the **thylakoid membranes** inside the chloroplast, where chlorophyll is found.
- Light energy is absorbed by chlorophyll.
- This energy is used to **split water molecules** into hydrogen and oxygen (photolysis).
- **Oxygen** is released as a waste product — this is where the oxygen we breathe comes from.
- The hydrogen is transferred to the second stage.

Stage 2 — The Light-Independent Stage

- Takes place in the **stroma** (the fluid surrounding the thylakoids).
- **Carbon dioxide** combines with the **hydrogen** from stage 1 to make **glucose**.
- This stage does not directly require light, but it depends on products made in stage 1.

Exam Tip: A common misconception: "light-independent" does *not* mean it happens in the dark. It still happens during the day in most plants — it just doesn't use light directly; it uses the hydrogen and energy produced in stage 1.

Stage	Location	What happens	Needs light?
Light-dependent	Thylakoid membrane	Water split → hydrogen + oxygen. Oxygen released.	Yes
Light-independent	Stroma	CO ₂ + hydrogen → glucose	No (uses products of stage 1)

6.3 Uses of Glucose in Plants

Glucose made in photosynthesis is used by the plant in several ways:

Use	Detail
Respiration	Glucose is broken down in mitochondria to release energy for the plant's own cells (growth, active transport of mineral ions in root hair cells, etc.).
Making cellulose	Glucose molecules are joined together to form cellulose, which is used to build plant cell walls — giving cells strength and shape.
Stored as starch	Excess glucose is converted to starch (a large, insoluble polymer) and stored in leaves and in storage organs such as tubers (potatoes). Starch is compact, insoluble, so it does not affect water balance.
Making amino acids (proteins)	Glucose + nitrate ions (absorbed by roots) → amino acids, which join together to form proteins.
Making lipids (oils)	Glucose can be converted to fats/oils, often stored in seeds (e.g. sunflower, rapeseed) as a concentrated energy store for germination.

Exam Tip: Remember: starch is made of glucose, cellulose is made of glucose, *proteins are not* — proteins need **nitrate ions from the soil** as well as glucose. This is why gardeners add nitrogen-containing fertilisers.

6.4 Limiting Factors of Photosynthesis

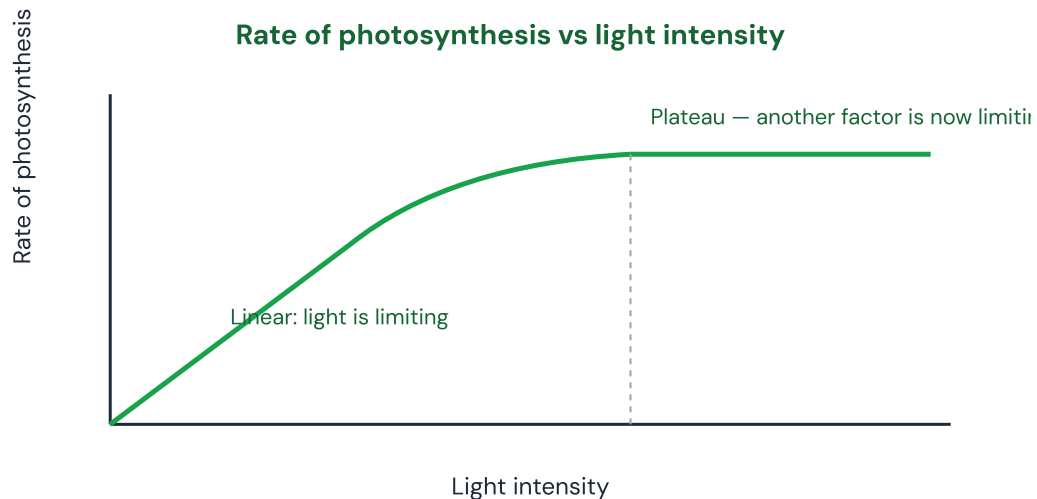
A **limiting factor** is the factor that, at a particular time, stops the rate of photosynthesis from increasing further. If you provide more of that factor, the rate will go up. If any other factor is the limiting one, adding more of this one will not help.

The three main limiting factors you must know are:

1. **Light intensity** — provides the energy for the light-dependent stage.
2. **Carbon dioxide concentration** — the raw material joined to hydrogen to form glucose.
3. **Temperature** — affects the enzymes that catalyse the reactions.

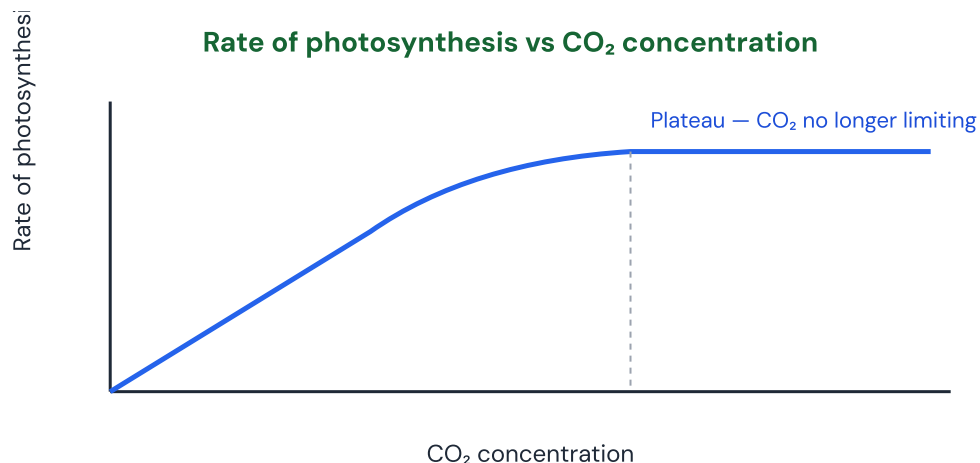
(a) Rate vs Light Intensity

As light intensity increases, the rate of photosynthesis increases proportionally — until something else becomes the limiting factor. The graph then **levels off (plateaus)**: light is no longer limiting.



(b) Rate vs CO₂ Concentration

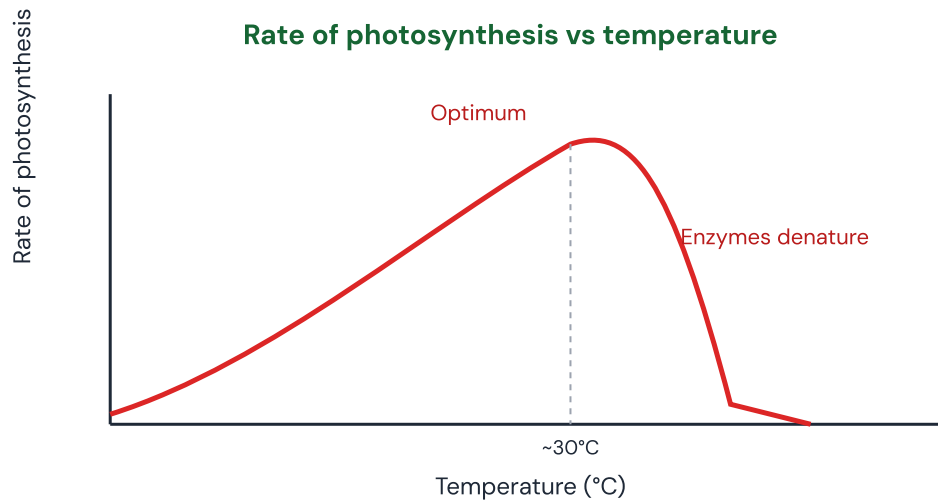
Identical shape — rate rises as CO₂ rises, then plateaus when something else (light or temperature) becomes limiting. Atmospheric CO₂ (~0.04 %) is usually the limiting factor outdoors on a bright, warm day.



(c) Rate vs Temperature

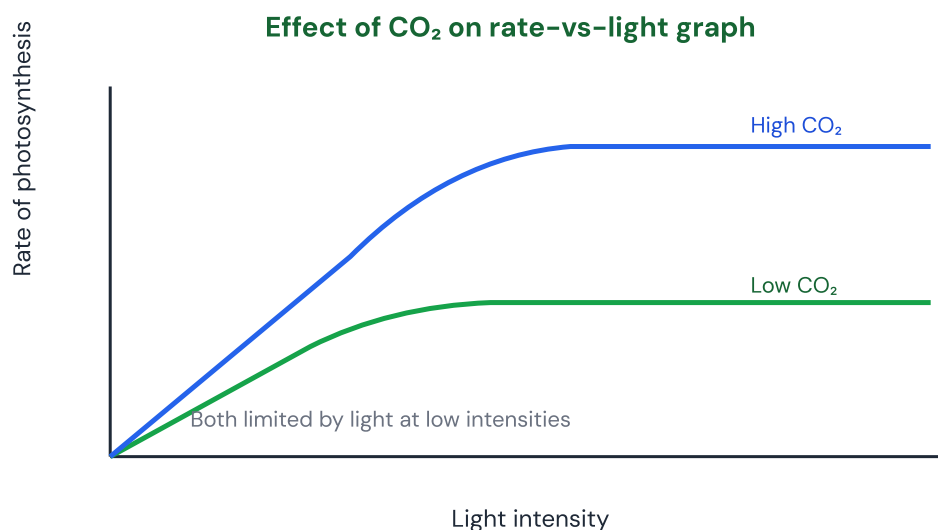
Temperature is different: rate increases with temperature up to an **optimum** (typically ~25–35 °C for most plants), then drops **sharply**. This is because the enzymes that catalyse

photosynthesis **denature** at high temperatures — their active sites change shape and substrates no longer fit.



Interaction: raising one factor shifts the plateau of another

If a graph is plateauing, increasing the *limiting* factor will make the plateau rise. For example, at high light intensity but low CO₂, adding more CO₂ will cause the graph of rate-vs-light to plateau at a higher rate.



Exam Tip: Read graphs carefully. "Light is limiting" = rising part; "something else is limiting" = flat plateau. You should be able to identify the limiting factor at any point on the graph.

6.5 The Inverse Square Law for Light Intensity

Light intensity from a point source falls off with the square of the distance from the source:

Light intensity $\propto 1/d^2$, where d is the distance from the light source.

This means if you **double** the distance, the light intensity falls to **one quarter**. If you **triple** the distance, it falls to **one ninth**. This is important for practicals (e.g. pondweed + lamp experiments) — you must keep the lamp at a carefully measured distance.

Distance d	$1/d^2$	Relative intensity
10 cm	$1/100$	1.00 (reference)
20 cm	$1/400$	0.25 ($\frac{1}{4}$)
30 cm	$1/900$	0.11 ($\approx \frac{1}{9}$)
40 cm	$1/1600$	0.06 ($\approx 1/16$)

Worked example

A student puts a lamp 10 cm from a beaker of pondweed and counts 40 bubbles of oxygen per minute. They move the lamp to 20 cm. How many bubbles per minute would they predict?

At 10 cm: $1/d^2 = 1/100 = 0.01 \rightarrow 40$ bubbles

At 20 cm: $1/d^2 = 1/400 = 0.0025 \rightarrow$ intensity is $\frac{1}{4}$

Predicted rate = $40 \times \frac{1}{4} = 10$ bubbles per minute

Exam Tip: Exam calculators often ask "calculate the light intensity" — just work out $1/d^2$ in consistent units. Units of light intensity in these questions are usually **arbitrary units (a.u.)**.

6.6 Using Limiting Factors in Greenhouses

In commercial horticulture, growers actively **remove** each limiting factor to maximise yield inside a greenhouse (glasshouse). The bigger/better the crop, the more money they earn — but each intervention costs money, so there is always a **commercial trade-off**.

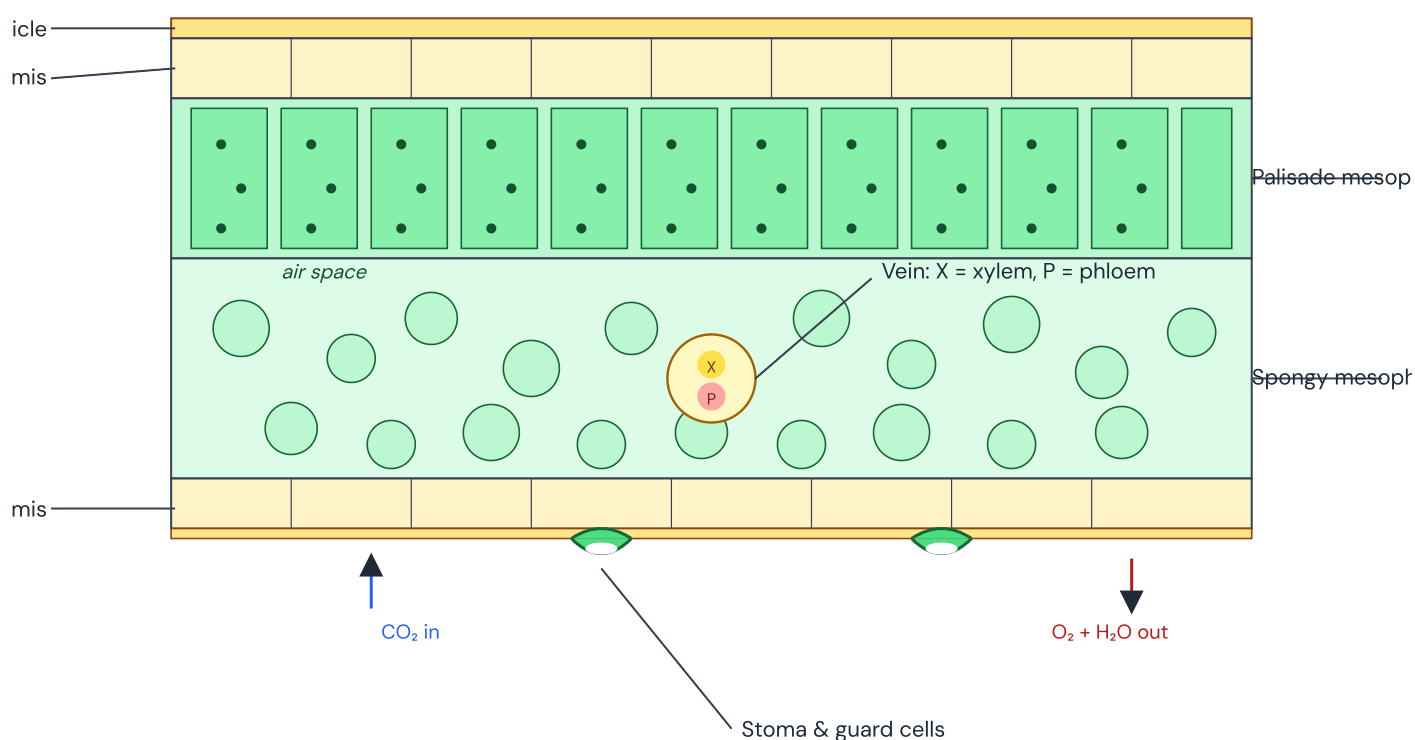
Factor	How it is controlled	Benefit	Cost / trade-off
Light intensity	Artificial lamps (LEDs, sodium lamps) to extend the day in winter	Faster growth; year-round production	Electricity is expensive; lamps give off heat and need replacing
CO ₂ concentration	Burn fuel (natural gas, paraffin) or pump CO ₂ from cylinders inside the greenhouse	Removes CO ₂ as the limiting factor — can roughly double growth rates	Fuel / CO ₂ costs; risk of CO if heaters malfunction
Temperature	Heaters in cold weather; ventilation and shading in hot weather	Keeps plants at enzyme optimum all year	Heating is the biggest running cost
Shading	Blinds, whitewash on glass, or movable screens	Prevents overheating and enzyme denaturation in summer	Reduces light intensity — use only when temperature is the problem
Water	Automatic irrigation / drip feed	No wilting; consistent supply	Equipment cost; risk of over-watering / root rot
Minerals	Fertilisers added to soil or hydroponic solution	Supplies nitrate, phosphate etc. for protein/DNA synthesis	Cost; environmental run-off

Growers decide which limiting factor is worth removing based on whether the extra yield and selling price outweigh the extra costs. For very high-value crops (tomatoes, strawberries, cannabis, ornamentals), full climate control is profitable. For low-value bulk crops, it is not.

6.7 Leaf Structure

A leaf is an organ perfectly adapted as the plant's main site of photosynthesis and gas exchange. It is thin and flat, with a huge surface area for absorbing light and CO_2 , and a network of vascular tissue to deliver water and remove sugars.

Leaf cross-section



How each layer is adapted

Layer	Feature	Adaptation / Function
Waxy cuticle	Transparent, waterproof wax on top surface	Lets light through; reduces water loss by evaporation; barrier against pathogens

Upper epidermis	Single layer of transparent cells — no chloroplasts	Lets light pass straight through to the palisade layer below
Palisade mesophyll	Long, column-shaped cells packed with many chloroplasts, near the top surface	Main site of photosynthesis — maximises light absorption
Spongy mesophyll	Irregular cells with large air spaces between them; some chloroplasts	Air spaces allow CO ₂ to diffuse to photosynthesising cells and O ₂ / water vapour to diffuse out
Vascular bundle (vein)	Xylem vessels (top) and phloem (bottom)	Xylem brings water & mineral ions; phloem takes sugars away
Lower epidermis	Contains stomata (pores) controlled by guard cells	Controls gas exchange & water loss
Stomata & guard cells	Pairs of kidney-shaped cells forming a pore	Open in the day for CO ₂ in / O ₂ out; close at night or in drought to save water

Exam Tip: The leaf has **many adaptations to be a good exchange surface**: large SA:V (thin and flat), short diffusion distance (one-cell-thick wall to each mesophyll cell), and a steep concentration gradient (photosynthesis continually uses up CO₂).

6.8 Stomata and Guard Cells

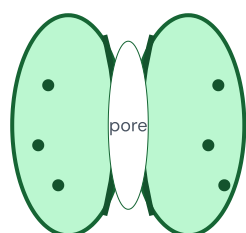
Each stoma (plural: stomata) is a pore in the lower epidermis surrounded by two **guard cells**. Guard cells have **unequally thickened walls** — the inner wall facing the pore is thicker than the outer wall. When water enters by osmosis, they become turgid and curve outwards, pulling the pore **open**. When water leaves, they become flaccid and the pore **closes**.

Condition	Stomata	Why
Daytime, plant well-watered	Open	CO ₂ needed for photosynthesis; O ₂ released

Night-time	Closed	No photosynthesis — closing reduces water loss
Hot / dry / windy	Closed (wilt)	Reduces transpiration to prevent wilting

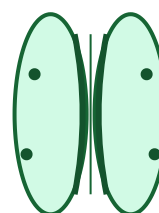
Stoma: open (day) vs closed (night / drought)

Open (turgid)



Guard cells swollen with water
— curve outwards, pore opens

Closed (flaccid)



Guard cells lose water
— become flaccid, pore closes

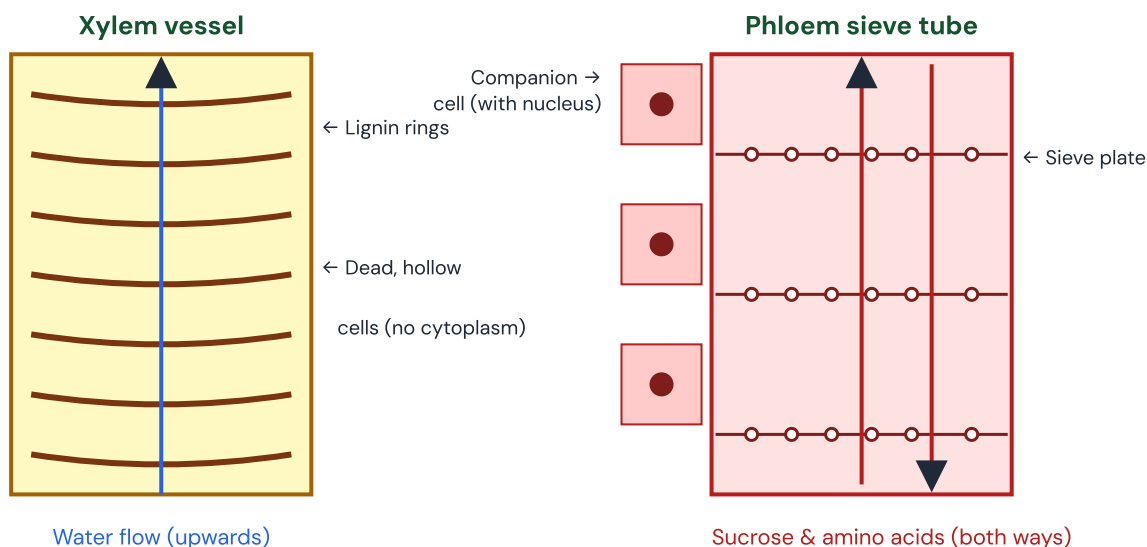
6.9 Xylem and Phloem — Transport Tissues

Plants have two specialised vascular tissues that work together to move substances around the plant. Xylem carries **water** upwards only. Phloem carries **dissolved food** (sucrose and amino acids) in either direction.

Feature	Xylem	Phloem
What is transported	Water and dissolved mineral ions	Sucrose and amino acids (assimilates, made in photosynthesis)
Direction	One way: roots → leaves (upwards only)	Both ways: from leaves (source) to growing/storage tissues (sinks)
Process name	Transpiration stream	Translocation
Cell structure	Dead, hollow, no end walls — a continuous tube. Walls reinforced with lignin (often in	Living elongated cells joined end-to-end by sieve plates (perforated end walls). Each sieve tube cell has a companion cell

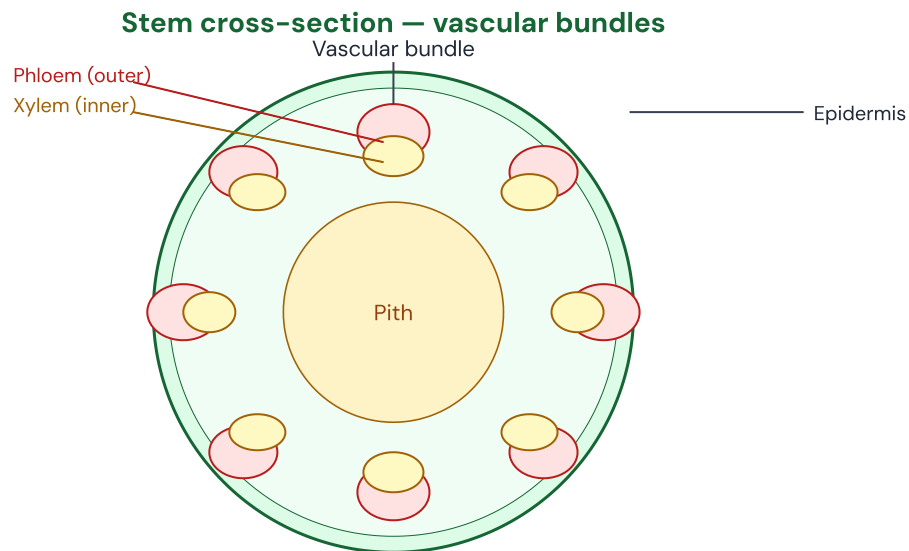
	spirals or rings) for support and waterproofing.	attached, containing a nucleus and many mitochondria (provides ATP).
Requires energy?	No — water is pulled up by transpiration (evaporation from leaves)	Yes — active transport of sucrose into sieve tubes requires ATP

Xylem and phloem structure



Position in the stem and leaf

In a young dicotyledonous stem, xylem and phloem are arranged in **vascular bundles** around the edge, with xylem on the inside and phloem on the outside. In a leaf vein, xylem is on the upper side and phloem on the lower side.



6.10 Transpiration

Transpiration is the loss of water vapour from a plant's leaves by **evaporation** through the stomata.

How the transpiration stream works

1. Water evaporates from the surfaces of the **spongy mesophyll cells** into the air spaces inside the leaf.
2. This water vapour diffuses out through the **stomata** into the surrounding air.
3. This pulls more water out of the xylem in the leaf vein (replacing the lost water).
4. A **continuous column of water** is pulled up the xylem from the roots — held together by hydrogen bonds between water molecules (cohesion) and sticking to the xylem walls (adhesion).
5. Water is absorbed from the soil by the **root hair cells** via osmosis to replace what was lost.

Pathway: **Soil → root hair cells → root cortex → xylem → stem → leaf → mesophyll → air spaces → stomata → air.**

Why transpiration is useful to the plant

- **Transport of mineral ions** from the roots to the leaves (ions dissolved in water move with the transpiration stream).
- **Cooling** — evaporation removes heat from leaves on hot days (like sweating).
- **Turgor** — water pressure inside plant cells keeps stems upright and leaves spread out; needed for growth by cell elongation.
- Delivers water to leaves for photosynthesis.

6.11 Factors Affecting Transpiration Rate — The Potometer

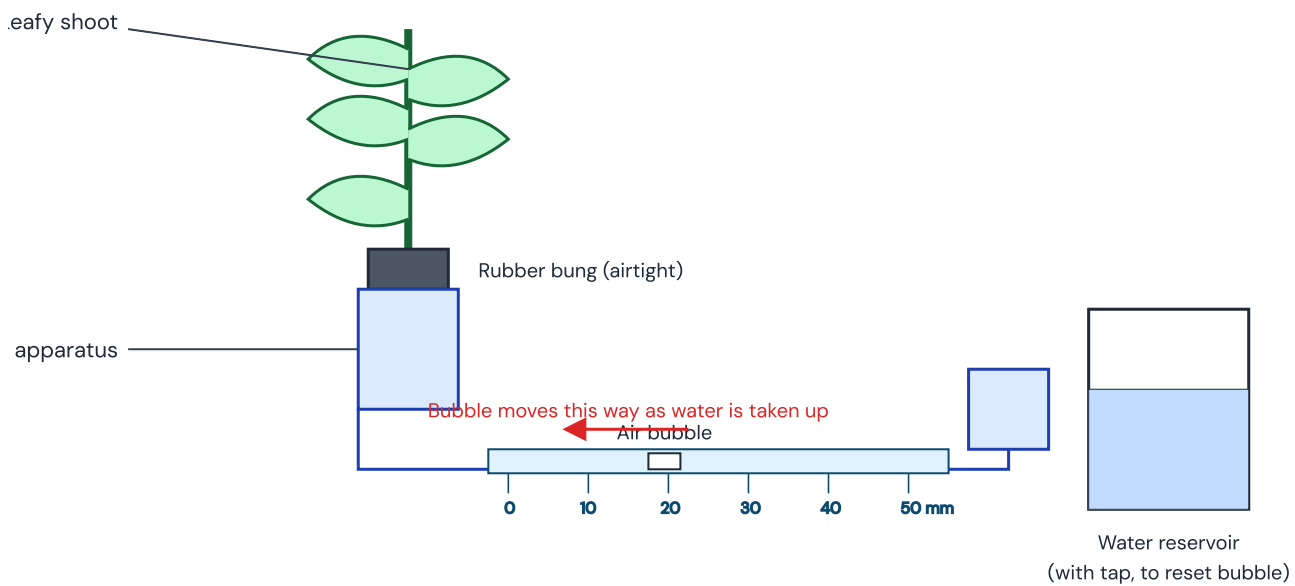
Transpiration rate depends on how fast water evaporates from the leaves. Four main factors affect it:

Factor	Effect on rate	Explanation
Light intensity	↑ light, ↑ rate	More light → stomata open wider → more water escapes
Temperature	↑ temperature, ↑ rate	Water molecules have more kinetic energy → faster evaporation and faster diffusion out
Air movement (wind)	↑ wind, ↑ rate	Wind removes humid air next to the leaf, keeping the water-vapour concentration gradient steep
Humidity	↑ humidity, ↓ rate	Less-steep concentration gradient — less water vapour diffuses out

Measuring transpiration with a potometer

A **potometer** measures the uptake of water by a cut shoot (used as a proxy for transpiration rate — almost all water taken up is lost by transpiration). The rate is measured as the distance moved by an air bubble in a capillary tube per unit time.

Potometer apparatus



Method (outline)

1. Cut a leafy shoot underwater (prevents air entering the xylem) and attach it to the potometer.
2. Ensure the apparatus is airtight and bubble-free except for one controlled air bubble in the capillary tube.
3. Record the starting position of the bubble.
4. Time how long the bubble takes to travel a known distance (e.g. 30 mm).
5. Change one variable (e.g. use a hairdryer/fan for wind; lamp for light; kettle for humidity). Keep all others constant.
6. Reset the bubble using the reservoir tap; repeat for each condition; calculate means.

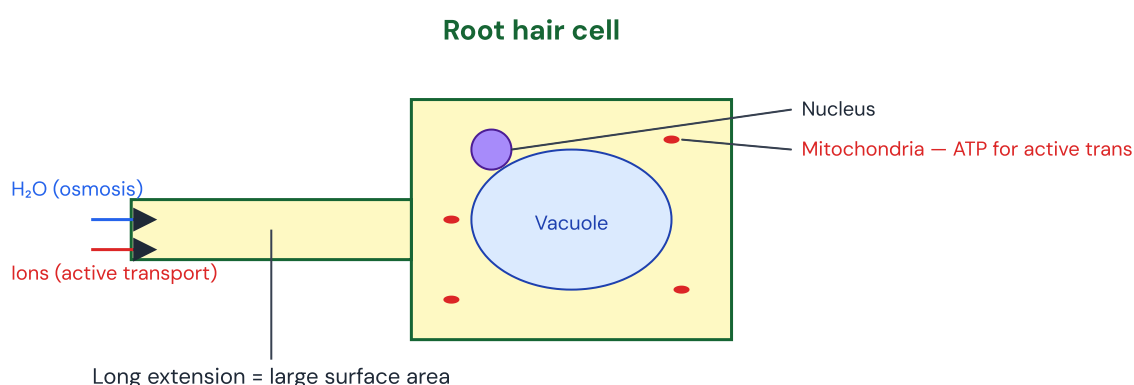
Exam Tip: A potometer measures *water uptake*, not transpiration directly. It assumes almost all water taken up is lost by transpiration — a small amount is used in photosynthesis and to maintain turgor, but this is negligible in short experiments.

6.12 Root Hair Cells

Water and mineral ions enter a plant through specialised cells called **root hair cells**, found just behind the root tip.

Adaptation	Function
Long, thin extension (the "hair")	Huge increase in surface area for absorbing water and mineral ions
Thin cell wall and membrane	Short diffusion / osmosis pathway
No chloroplasts (underground — no light)	Space saved for absorption machinery
Large permanent vacuole with cell sap	Keeps cell contents more concentrated than soil water — water enters by osmosis
Many mitochondria	Release ATP for active transport of mineral ions (against the concentration gradient)

Water enters by **osmosis** (soil water is more dilute than cell sap). Mineral ions (nitrate, phosphate, potassium, magnesium) enter by **active transport** because they are usually in lower concentration in the soil than inside the cell.



6.13 Plant Hormones — Auxins

Auxins are plant hormones produced in the **tips** of growing shoots and roots. They move down from the tip to the **zone of elongation**, where they cause cells to elongate (grow

longer).

Key facts about auxin:

- Made at the tip (apex) of shoots and roots.
- Moves down from the tip in a concentration gradient.
- Causes **cell elongation** in the region just below the tip.
- Its effect depends on concentration **and** tissue type:
 - **In shoots:** higher auxin concentration → more elongation.
 - **In roots:** higher auxin concentration *inhibits* elongation (the root cells grow less, not more).

6.14 Tropisms — Phototropism and Gravitropism

A **tropism** is a directional growth response to a stimulus. "Positive" = growth toward the stimulus; "negative" = growth away from it.

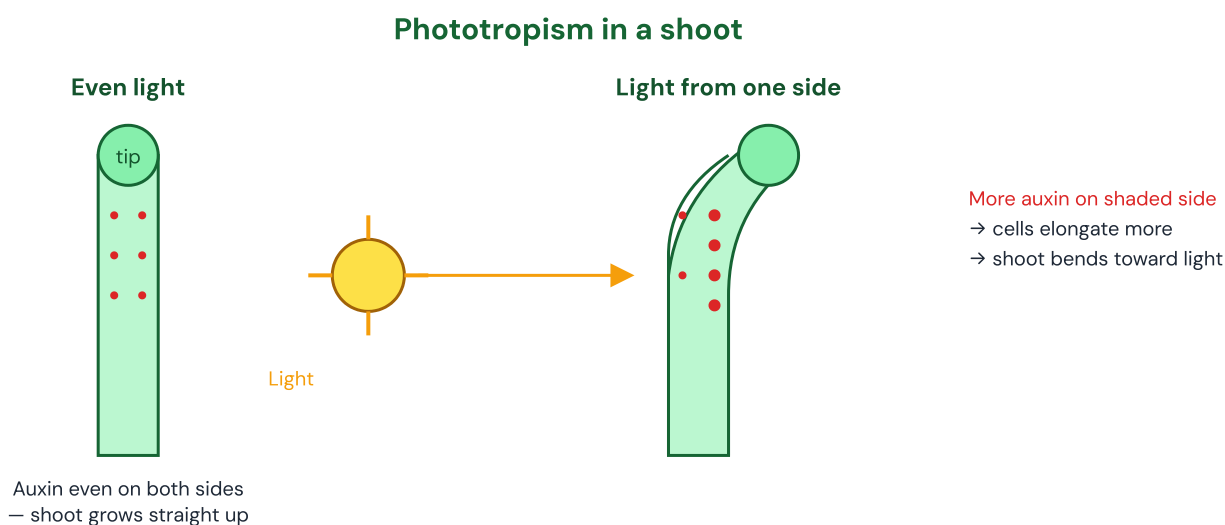
Tropism	Stimulus	Shoots	Roots
Phototropism	Light	Positive — grow toward light	Negative — grow away from light
Gravitropism (geotropism)	Gravity	Negative — grow upwards (against gravity)	Positive — grow downwards (with gravity)

How auxin causes phototropism (shoot bending toward light)

1. Light hits the shoot tip from one side.
2. Auxin is redistributed — it moves to the **shaded** side of the shoot.
3. More auxin on the shaded side → cells on that side elongate *more*.
4. The shaded side grows longer than the lit side — the shoot **bends towards the light**.

How auxin causes gravitropism in roots

1. When a root is horizontal, auxin accumulates on the **lower** side due to gravity.
2. In roots, high auxin *inhibits* cell elongation.
3. So the lower side elongates less than the upper side — the root **bends downwards** (positively gravitropic).
4. In shoots, the same redistribution of auxin to the lower side *stimulates* elongation there, so shoots bend upwards (negatively gravitropic).



6.15 Uses of Plant Hormones in Agriculture

Plant hormones can be extracted, made synthetically, and applied to crops to control growth. You need to know three groups of hormones and their commercial uses.

Auxins — uses

Use	How it works
Rooting powder	Cut stems dipped in auxin powder grow roots much faster — used to produce clones of prize plants from cuttings.
Selective weedkillers (e.g. 2,4-D)	High doses of synthetic auxin cause broad-leaved plants (weeds like dandelions) to grow abnormally fast and die. Narrow-leaved

	cereals (wheat, barley) absorb less and survive — the crops are left intact.
Tissue culture / micropropagation	Adding auxin (with cytokinin) to growing media stimulates cells in callus tissue to grow into new roots and shoots, producing many genetically identical plants.

Gibberellins — uses

Use	How it works
End seed dormancy / promote germination	Applying gibberellin breaks dormancy and triggers germination all at once — useful for uniform crop emergence.
Induce flowering	Makes plants flower earlier, or flower when they otherwise would not.
Grow larger fruit	Spraying gibberellin on seedless grapes, for example, makes them grow bigger and juicier.

Ethene — uses

Use	How it works
Ripening fruit during transport / storage	Fruit (e.g. bananas, tomatoes) are picked green and unripe so they do not spoil. Before sale, they are exposed to ethene gas, which triggers ripening — fruit starts to produce its own ethene too, so ripening spreads.

Exam Tip: A classic exam question: "Why does placing a ripe banana next to unripe pears speed up their ripening?" Answer: ripe fruit release **ethene**, which triggers the enzymes that ripen other nearby fruit.

Summary — Key Equations & Relationships

Concept	Equation / Rule
Photosynthesis	$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (light, chlorophyll; endothermic)
Inverse square law	Light intensity $\propto 1 / d^2$
Rate of reaction (practicals)	Rate = 1 / time, or bubbles per minute, or cm^3 of O_2 per minute
Transpiration pathway	Root hair $\rightarrow$ cortex $\rightarrow$ xylem $\rightarrow$ leaf $\rightarrow$ stomata $\rightarrow$ air
Translocation (phloem)	Source (leaves) $\rightarrow$ Sink (roots / fruits / growing tips); both up & down

Key Definitions — Glossary

Term	Definition
Photosynthesis	Endothermic reaction in which plants use light energy to convert CO_2 and water into glucose and oxygen.
Chlorophyll	Green pigment in chloroplasts that absorbs light energy for photosynthesis.
Endothermic	A reaction that takes in energy from the surroundings.
Limiting factor	The factor currently preventing the rate of a process from increasing further.
Inverse square law	Light intensity from a point source is proportional to $1/d^2$, where d is the distance from the source.
Transpiration	Loss of water vapour from leaves by evaporation through stomata.
Translocation	Transport of sucrose and amino acids in phloem from leaves (source) to other parts of the plant (sinks).

Stoma	A pore in the lower epidermis of a leaf, flanked by two guard cells, that controls gas exchange and water loss.
Xylem	Dead, hollow, lignified tissue that transports water and dissolved mineral ions up from roots to leaves.
Phloem	Living tissue of sieve tubes (with sieve plates) and companion cells that transports sucrose and amino acids both up and down.
Tropism	Directional growth of a plant in response to a stimulus such as light or gravity.
Auxin	A plant hormone produced in shoot and root tips that controls cell elongation.
Gibberellin	A plant hormone that promotes seed germination, flowering and fruit growth.
Ethene	A gaseous plant hormone that triggers fruit ripening.
Potometer	An apparatus used to measure the rate of water uptake (and hence transpiration) by a cut leafy shoot.